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Channel-scale optimisation and tuning of large tidal turbine arrays using LES with adaptive mesh

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ABSTRACT

Large arrays of tidal turbines are critical to realise the potential of tidal current power. This study is a systematic exploration of large tidal array optimisation in channels with numerically modelled array layouts in 2-D. Crucially, flow along channels is driven by head loss leading to significantly more realistic results than previous models which assume constant velocity. The 2-D adaptive mesh approach bridges the gap between large- and small-scale array models. Hundreds of layouts and turbine tunings have been simulated using LES of turbulent flow in tidally reversing currents to explore channel-scale optimisation and tuning of large arrays. Simulations show that total power capture increases as rows are added to the array although there are diminishing returns on additional turbines. Each turbine in 1 (7), optimally blocked row in a small channel captures 2.5× (0.5×) the power of an isolated turbine. There is an optimum blockage for maximum power per turbine which decreases linearly from 1.0 as the number of rows increases. As array size increases individual turbine wakes become less important than stepped head loss across each row. Free-stream velocity reduces linearly with total power capture, with the gradient increasing with channel size.

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1. Introduction

In order to capture an economically significant amount of energy, arrays of tidal turbines will need to increase in size from present test deployments. Once these arrays reach a large size, they change the large-scale flow characteristics of the channel they are deployed in. Most previous studies of power capture in arrays implicitly apply to small arrays because they impose a fixed free-stream velocity at open boundaries. Garret and Cummins developed an analytical model that assesses a turbine in a channel [1]. This analytical model was extended to apply to a single, small, packed row of turbines in a wide channel by Ref. [2]. Small arrays of idealised turbines have previously been modelled in 2-D or 3-D using a range of turbine representations. Idealised drag elements, Blade Element Momentum Theory, fixed rotor, rotating line element and fixed or rotating blade models have all been applied to small arrays (e.g.: [3–7], respectively). These studies describe wake deficits or wake shadows. However, they do not discuss the larger, channel-scale free-stream velocity reduction or head loss across turbines, or arrays of turbines. The optimum layouts derived from these models are only applicable to arrays with small numbers of turbines because they assume that the free-stream velocity in the channel is not reduced by the addition of the array.

Once an array is large enough to reduce the channel's free-stream velocity by even a small fraction, the optimum array layout for maximum power capture can only be modelled if the flow is driven by a tidal head loss between the ends of the channel, [8–11]. In the present study a pressure head loss is used to represent the tidally-oscillating head loss along the channel in a 2-D array-scale model (with reference to individual turbine wakes). Driving the flow with a head loss in this way is crucial to understanding the dynamics of large arrays and their impact on natural flow.

Vennell has previously used a tidal head loss to drive the flow in a 1-D analytical model of array layout optimisation in a channel [8,12–14]. In Vennell's studies, Garrett and Cummins' channel model [1] was combined with their earlier turbine model [15].

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These studies simulate rows of turbines with no representation of individual turbines or their wakes. Adcock et al. also simulated rows of turbines rather than individual turbines, in a 2-D hydrodynamic model of Pentland Firth [16]. Refs. [16,17] assess the power capture as basin efficiency where the power available to the turbines is expressed as a fraction of the available power in the channel. The understanding of optimised large array layouts has been limited by models/studies that do not incorporate individual turbines with flow driven by a head loss. Further, turbines in an array should be tuned to have the optimum thrust or drag coefficient for maximum power capture by the array over a tidal cycle in a given array layout [12].

The present study models individual, tuned turbines and their wakes in an exploration of power capture by different array layouts. This paper provides a systematic study of optimising array layout for large arrays of uniform rows in two tidal channels using a 2-D hydrodynamic model. The aim of the model development is to capture the *channel-scale* dynamics of fluid flow around arrays of tidal turbines within a channel. In order to achieve this certain adjustments are made to the model; neither the full turbulent structure in the wake of individual turbines, nor the large scale complexity of full tidal flow in a channel with real bathymetry are captured. It is the channel- and array-scale flow that is modelled with sufficient detail to compare the power removal by individual turbines in different array layouts. Driving the flow through the channel with a tidal head loss rather than a given velocity is fundamental to understanding power extraction and the underlying hydrodynamics that drive this power capture by large arrays. The present model is used to compare estimates of the power removed from the flow by each of a large number of layouts of optimally tuned tidal turbine arrays and to find optimum layouts for maximum power capture. Further, two linear relationships i) between free-stream velocity reduction in the channel and the total power captured by the array and ii) between optimal global blockage for maximum power capture and number of rows in the array are identified.

2. Model methodology

Supplementary video related to this article can be found at http://dx.doi.org/10.1016/j.renene.2015.09.048.

The adaptive mesh capability of the Gerris flow solver [18,19] is used to model tidal arrays where the scales range from sub-turbine scale (metres) to channel-scale (kilometres). Gerris solves the 2-D incompressible Euler equations for the flow through an idealised channel containing arrays of uniformly spread rows of turbines of different blockages to compare the power capture from arrays with different layouts. The equations solved are

$$\frac{\partial \mathbf{U}}{\partial t} = -\frac{\nabla p}{\rho} - \mathbf{U}\nabla\mathbf{U} \tag{1}$$

and

$$\nabla\cdot\mathbf{U} = 0 \tag{2}$$

where $\mathbf{U}(x,y,t) = (U(x,y,t), V(x,y,t))$, U and V are the along-stream and across-stream components of velocity respectively, and $p = p(x,y,t)$ is the pressure field at location (x,y) and time t. The free surface elevation is incorporated into the pressure term. At the channel-scale Coriolis is assumed to be insignificant. Gerris solves Equations (1) and (2) for $\mathbf{U}$ and p using a second-order, unconditionally stable Godunov technique to solve the advection term [20]. The resulting solutions are second-order accurate in space and time.

Friction due to turbines is applied within cells that are occupied by a turbine (turbine friction C_t within area A_t) while depth averaged natural bottom friction (C_d/h) is applied throughout the channel using the oceanographic definition of depth-averaged bed shear stress,

$$F = -\rho\left(A_t C_t + \frac{C_d}{h}\right)\mathbf{U}\left|\mathbf{U}\right| \tag{3}$$

where C_t and C_d are coefficients of friction and h is the mean water depth in the channel. Friction is applied by adjusting U and V by decelerating the flow at the end of the numerical time-step using $F = \partial U/\partial t$ through

$$U\left(t + \frac{1}{2}\Delta t\right) = \frac{U(t)}{\left(1 + dt|\mathbf{U}(t)|\left(\frac{C_d}{h} + A_t C_t\right)\right)} \tag{4}$$

and

$$V\left(t + \frac{1}{2}\Delta t\right) = \frac{V(t)}{\left(1 + dt\left|\mathbf{U}(t)\right|\left(\frac{C_d}{h} + A_t C_t + C_S A_S\right)\right)}, \tag{5}$$

in a standard approach e.g.: [19] and [10]. $C_S = 20$ describes a damping force to across stream velocity applied in the plan area, A_S, $60\Delta x_{min}$ inside either end of the channel to damp unwanted oscillations in across stream flow at the inflow end on each reversing tide (Fig. 1a).

This technique is applied here to Gerris' 2-D Navier Stokes solver whereas [10,19] applied bottom friction in this way using Gerris' GfsOcean solver. The equations solved by the GfsOcean solver assume that there are no length scales shorter than the water depth. This assumption is not strictly true when turbines are smaller than the water depth. No such shallow water assumption is made in the 2-D Navier Stokes solver used in the present study.

2.1. Uniform rows of turbines

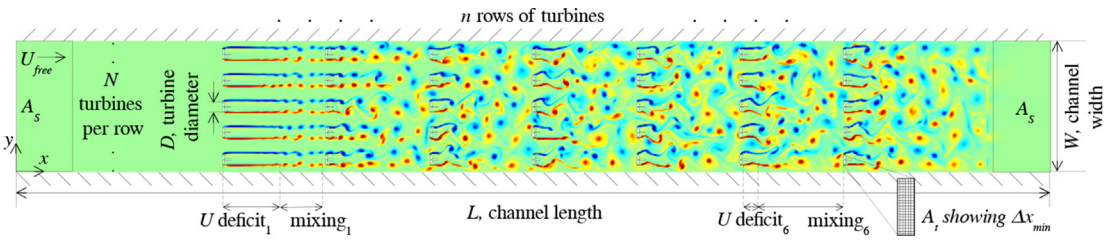
Uniform arrays are constructed of rows of turbines that are uniformly spread across the width of a channel, following Garrett and Cummins and Vennell [8,15]. There are N turbines with diameter D in each row where each row is $10D$ apart from the next, Fig. 1a. The turbines are evenly spaced across the channel so that the hub to hub spacing is W/N, and are assumed to fill the depth of the channel, so the global blockage,

$$\varepsilon = ND/W. \tag{6}$$

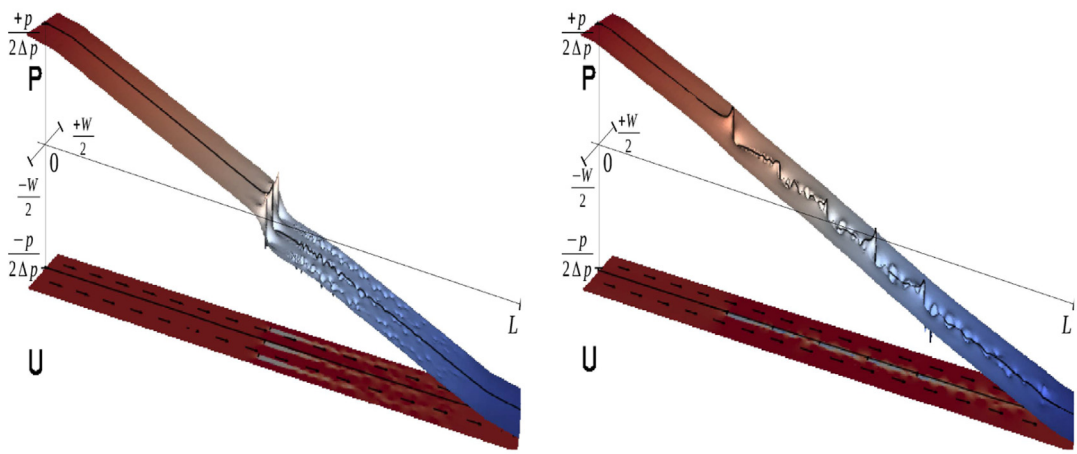
Different global blockages are achieved by changing the number of turbines in a row. As extra turbines are added to increase the global blockage, the spacing between turbines reduces until the spacing is the smallest cell size at $\varepsilon = 0.96$.

2.2. Adaptive mesh with monotonic increasing large eddy simulation

Gerris' adaptive mesh capability allows the cell size and time step to change depending on a refinement criterion. This saves simulation time by focussing computational effort on regions of the simulation where the flow is more dynamic. An initial grid with minimum cell size of $\Delta x_{min} = D/20$ around the turbines and up to $\Delta x_{max} = 125\Delta x_{min}$ elsewhere was set up ($\Delta x_{min} = W/250$ in the small channel, $\Delta x_{min} = W/1000$ in the large strait). The grid is refined every ten time steps using vorticity refinement criteria, so that the cell size reduces in areas of high vorticity, down to the

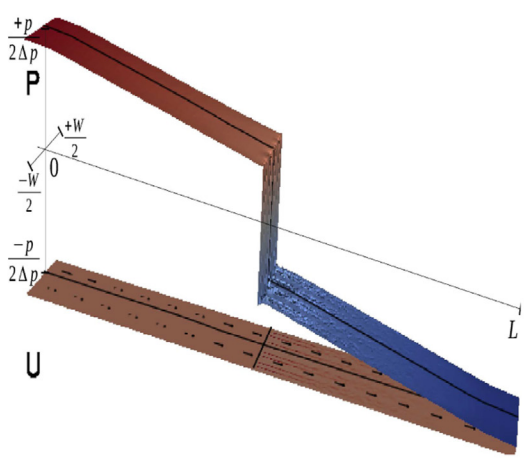


(a) Colour shows vorticity (dark red clockwise, dark blue anti-clockwise) in the flow field around an array of 7 rows of 5 turbines optimally tuned for maximum power extraction by this layout in the small channel.

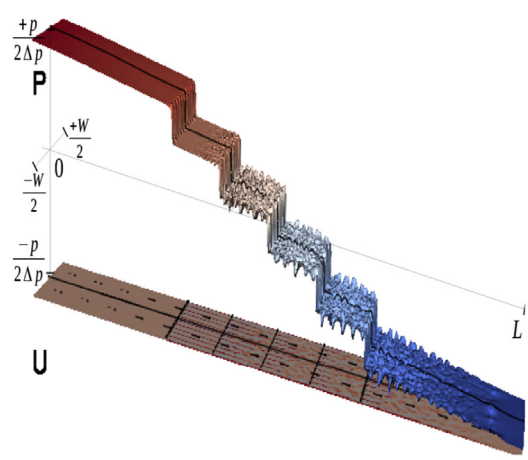


(b) 1 row of 3 turbines.

(c) 5 rows of 1 turbine per row.



(d) 1 row of 12 turbines.



(e) 5 rows of 10 turbines per row.

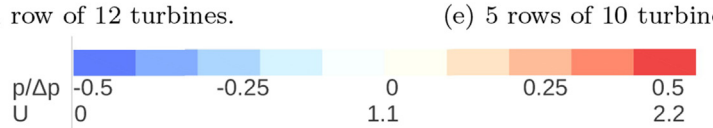


Fig. 1. a) Plan view of the channel and array layout. b to e) The instantaneous head loss (top, vertical scale and colour indicate normalised pressure) and along-stream velocity (bottom, direction and magnitude indicated by colour and arrows, turbine location indicated by black lines) in a channel around example array layouts in the small channel. Flow is maximum towards the right. Movies showing the tidally oscillating pressure head and velocity are in Ref. [11] (e.g.: ESM Movie 1). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

smallest specified size of Δx_{min}, and increases in areas of low vorticity, up to the maximum allowable size of $16\Delta x_{min}$ everywhere other than in a small region near the ends of the channel. At the channel ends, a larger grid size $\Delta x = W/4$ is enforced to help straighten the inflow from the open boundaries after the tide reverses. In this way, the section of the channel upstream of the array, where the flow is uniform, is filled with large cells and the flow in the turbulent turbine wakes is filled with smaller cells that adapt in size based on the turbulent dynamics of the wake.

The goal of the adaptivity here is to resolve flow around eddies shed in turbine wakes to higher resolution. By adapting based on vorticity, regions of small cell size follow regions with increased

shear. Increased shear is indicative of smaller scale flow features so the numerical effort is focussed on these regions. The same vorticity criterion have successfully been used by [e.g.: [10,18–21].

The time step also adapts to satisfy the CFL criteria. Minimum time steps are roughly [0.1] s but this is not enforced, so some models may have adapted to smaller time steps for brief periods when there is particularly high local vorticity or fast current around eddies in the flow.

Gerris' advection scheme does not contain an explicit parameterisation for expressing sub-grid turbulence. However, there is numerical viscosity resulting from the higher order errors that occur with a discrete representation of the solution, as in any numerical scheme [20]. In these simulations, the Monotonic Increasing Large Eddy Simulation (MILES) approach to modelling sub-grid turbulence is used where it is assumed that the numerical viscosity describes the turbulent viscosity, accounting for sub-grid transfer of energy. Ref. [20] points out that [22,23] have shown that this numerical dissipation can approximate the sub-grid turbulent energy transfer as well as other approaches used to describe the viscosity in semi-empirical viscosity models. The depth-averaged natural bottom friction applied throughout the domain provides further eddy dissipation but is not an attempt to replicate sub-grid transfer of energy. The MILES approach possibly results in turbulent wakes that persist further downstream than they would with an increased eddy viscosity applied. Numerical consistency for this technique has been demonstrated through converging power capture for a single turbine with increasing cell size [10,24]. Further, Popinet and Rickard describe an extensive validation of Gerris' solver [18–21].

2.3. Rescaled small channel and large strait

A further computational cost saving was attempted by rescaling the M2 driving period, $2\pi/\omega$, to one-tenth of the physical period, attempting to simulate tidally varying current in one-tenth the time that would be required in a domain without this rescaling. This increased tidal frequency is balanced by an increased bottom friction and decreased channel length to keep dynamic similarity of two ratios: the balance between inertia and friction (Ref. [8]'s $\lambda_0 = \alpha C_d L/h$) and the ratio of tidal excursion to channel length (Ref. [13]'s $\alpha = g\zeta_0/\omega^2 L^2$) the same as two real channels, where L is the channel length. When $\lambda_0 > 1$ the channel's dynamics are dominated by friction, when $\lambda_0 < 1$ the channel is dominated by inertia. When $\alpha > 1$ the length of the channel is shorter than the tidal excursion or the horizontal distance a parcel of water travels in half a tidal cycle.

A "small channel" that has $\lambda_0 = 5$ and $\alpha = 1.84$, similar to a relatively small, shallow channel such as Tory Channel, New Zealand where friction is dominant. The "large strait" with $\lambda_0 = 0.5$ and $\alpha = 0.21$ has a dynamical balance dominated by inertia that is closer to Cook Strait. The dimensions for these channels with reduced period that were used in this study are shown in Table 1 with full scale equivalents and real channels for comparison. Channel width, $W = L/8$, was chosen fairly arbitrarily to be close to the aspect ratio of Tory Channel and to have reasonable range of blockages available with a few turbines. These two test cases are useful for calculating and comparing the power capture and the reduction in flow caused by large arrays in two channels with different dominating natural dynamics. Further description of the rescaling and demonstration of consistency with full tidal period is provided in Ref. [11].

2.4. Flow driven by a tidally reversing pressure gradient

The flow through the channel is, crucially, driven by a pressure head at either end of the channel, establishing a pressure gradient along the length of the channel. This is a 2-D representation of the free surface elevation that drives the flow through channels such as Cook Strait or Tory Channel, New Zealand. The pressure gradient is varied sinusoidally by Dirichlet boundary conditions for $p = 7.5cos(\omega t)$ at the left and $-p$ at the right end of the channel. These boundary conditions establish a pressure gradient along the length of the channel, allowing velocity within the channel to adjust to a value that is consistent with this driving head gradient and the added friction of each array layout, as shown in Fig. 1b–e. These boundary conditions also establish a static pressure of 0 at the centre of the channel, aiding visualisation. A zero normal derivative for the velocity, $dU/dx = 0$, is also applied at each end of the channel. Free-slip boundary conditions are applied at the channel walls. Velocity is initialised to zero everywhere.

2.5. Turbine drag is tuned for each array layout

Following the approach in Ref. [10], 2-D rectangles of increased bottom friction are used to represent each turbine. Each turbine is 20 cells in diameter at the maximum resolution of the adaptive grid in the across stream direction (y) by 6 cells in the along stream direction (x), Fig. 1a. The diameter is similar to the presently deployed MCT turbine which is 18 m in diameter. The turbine width was chosen fairly arbitrarily to be large enough to avoid numerical discontinuities that could arise from turbines that are one or two cells wide but not so wide as to be unrealistic. The turbine drag coefficient, C_t, within this box represents the total drag of the turbine due to support structure as well as the active blade area.

The instantaneous power extracted from the array is calculated by summing the force times velocity over the area of all turbines in the array by

$$Power = \sum_{turbine\ area} \rho C_t U_t^2 |U_t|, \tag{7}$$

which is summed over 1 tidal cycle for each array simulated using

$$Energy = \int_{T/2}^{3T/2} Power\ dt \tag{8}$$

to get the energy captured by the array in a tidal period, T. It is assumed that the power dissipated by support structures is small. The first half tidal cycle is discarded as in that time the phase of the velocity is adjusting to the forcing imposed by the array and a wake is established downstream of the array. Divett (2014) explored the run time required for model spin up and showed that half a tidal cycle was sufficient [11].

The optimum tuned drag coefficient, C_t^{opt}, for maximum power extraction, $P_{per\ turb}^{tuned}$ was found for each specific array layout by running several simulations with different C_t and plotting a tuning curve (presented in Ref. [11]) following the approach developed in the analytical model in Ref. [8]. A fifth order polynomial was fitted to each tuning curve as it qualitatively fitted the trend. Tuned turbines in a specific layout have$C_t = C_t^{opt}$ for that layout. The process of finding these peaks is referred to as tuning. The results presented here represent only values for optimally tuned turbines for maximum power capture in each individual array layout. $P_{per\ turb}^{tuned}$ as presented below has normalised by the power captured by a single turbine.

Table 1
The dimensions of the small channel and large strait used in this study in rescaled geometry simulated and the physical dimensions of dynamically similar channels, Fig. 1a. All simulations in this thesis are run in the small or large channel with reduced period.

Channel	$T = \frac{2\pi}{\omega}$ hr	λ_0	α	U_0 m s^{-1}	L km	$g\zeta_0$ m^2 s^{-2}	$C_d \times 10^{-3}$	h m
Small full-scale	12.4	5	1.84	2.2	20	14.54	5	37
Small reduced period	1.24	5	1.84	2.2	2	14.54	25	18.4
Large full-scale	12.4	0.5	0.21	2.2	80	27.18	5	172
Large reduced period	1.24	0.5	0.21	2.2	8	27.18	25	86
Pentland Firth[a]	12.4	1.6	1	2.5	23	12	5	70
Cook Strait[b]	12.4	0.35	0.3	1.9	50	14	2.5	100
Tory Channel[c]	12.4	5.06	3.9	2 to 3	16.8	22	3	39.2

[a] From Ref. [25].
[b] From Ref. [8].
[c] From Ref. [26]

3. Results

The power per turbine as well as the total power captured by the whole array of tuned turbines are compared here for each array layout of 1–7 rows. Further, the channel-scale flow characteristics and optimum turbine tuning are compared for each layout of tuned turbines. There are three inter-related components to the fluid dynamics that drive the optimum array layout for maximum power capture, where turbines in each layout have been tuned for maximum power capture:

- the average upstream velocity in the channel (free-stream velocity)
- the head loss across each turbine or row of turbines and
- the tuned turbine drag coefficient.

While the dynamical behaviour may vary between channels depending on their dynamical balance of friction and inertia, the fluid dynamics of a given channel can be explored with these three components.

3.1. Comparison with Vennell's 1-D analytical model

Vennell's 1-D analytical model has been applied here to the small channel with the same λ_0 and α used in the 2-D numerical simulations. The optimum layout and power capture from the two approaches are compared.

It is clear that the main predicted trends with increasing blockage and number of rows are the same between the two approaches despite some differences, Fig. 2a. As in the numerical model, each turbine in a single row captures more power from the flow than in multiple rows at each blockage. Further, both approaches show an optimal blockage, ε^{opt} for maximum $P^{tuned}_{per\ turb}$ at which each turbine captures the most power from the flow, for an array with n rows.

However, there are three differences between results of the two models. The power capture predicted by the analytical model for arrays $n > 1$ with low ε is 30%–50% less than the power capture calculated by the same layouts in the numerical model (Fig. 2a and b). In the numerical model, the power captured by the flow by two or more rows of one turbine ($\varepsilon = 0.08$) decreases steadily as rows are added. This reduced power capture in the numerical models is likely due to the wake interaction. This reduced power capture for low ε is not represented in the analytical model, where there is no mechanism to describe this wake interaction. This interaction may dominate the power potential of arrays with only 1 to 3 turbines per row uniformly spread in multiple rows.

Secondly, each ε^{opt}_{ana} predicted by the analytical model (pluses) is slightly lower than that predicted by the numerical model (crosses), Fig. 2a. This is possibly due to the difference in power captured at low ε for more than 1 row pulling the analytical curves in Fig. 2a towards lower ε.

Finally, the numerical model predicts 35% more power capture than the analytical model for a single row at high ε (Fig. 2a). It is still unclear exactly what effect is responsible for the 1-D model predicting more power capture by a single highly blocked row than the 2-D model. One further difference between the two models that may account for this is that the analytical model assumes that complete mixing takes place within the length of the channel.

3.2. Tuned power per turbine

The 2-D numerical results for the small channel show there is an optimum global blockage for maximum power per turbine for an array with a given number of rows of tuned turbines, Fig. 2a. Further, this ε^{opt} decreases linearly as the number of rows increases (Fig. 3a). Extending this to significantly more than 7 rows would eventually led to a non-physical $\varepsilon^{opt} \leq 0$. The decrease in the optimal global blockage for maximum power is from $\varepsilon^{opt} = 0.96$ for 1 row to $\varepsilon^{opt} = 0.4$ for 7 rows. This optimal global blockage is arrived at through a complex interaction between the added drag of each turbine, losses from turbulent mixing in the wakes and interaction between wakes. The peak is due to the balance between getting more power by adding more turbines at $\varepsilon < \varepsilon^{opt}$ and the power lost when the free-stream flow is slowed beyond optimum when $\varepsilon > \varepsilon^{opt}$.

For any given global blockage, turbines in 1 row capture significantly more power per turbine than turbines in an array with multiple rows. The power per turbine at this optimum global blockage decreases as the number of rows in an array increases, Fig. 3b. The decrease in power per turbine captured by 1–7 optimally blocked rows is 380% from (2.5× to 0.66× tuned power for a single turbine). Given that this is the maximum power that it is possible to capture from the flow by arrays optimised for global blockage and turbine tuning, it is clear that this represents a strong limitation on the efficiency of each turbine within an array. There might be some small percentage gains that could be made by optimising the positions of the turbines within each row [24] or tuning each turbine specifically for its location in the array [12]. However, at the channel-scale, it is vastly more efficient in terms of $P_{per\ turb}$ to completely block a single row than to spread turbines evenly across more rows, in agreement with [8,16].

3.2.1. ε^{opt} decreases linearly as rows are added to the array

The linear trend for the 2-D numerical simulations between ε^{opt} and n for 1 to 7 rows in Fig. 3a suggests that, at least in this range of n,

$$\varepsilon^{opt} = -0.10n + 1.06. \tag{9}$$

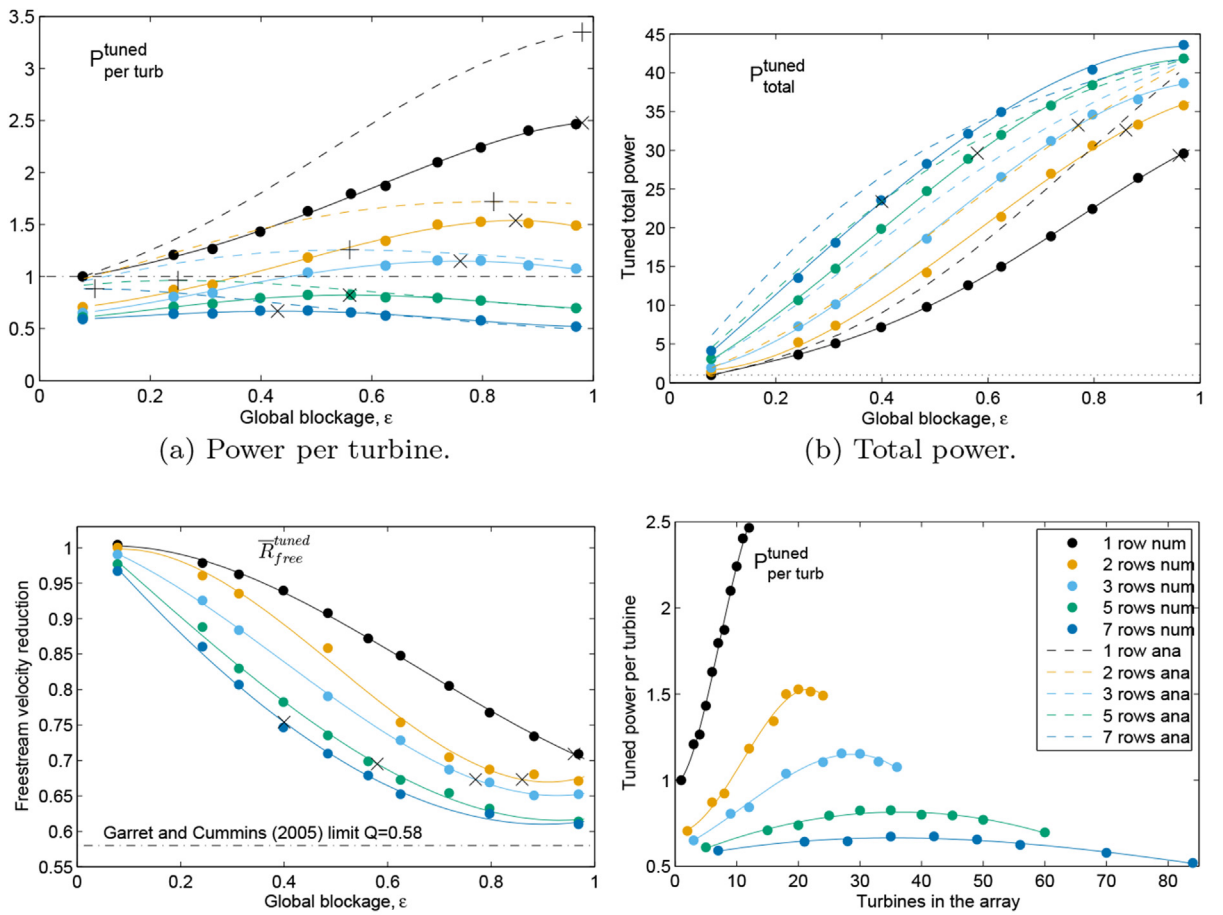


Fig. 2. a), b) and d) Power capture by arrays of uniform rows of optimally-tuned turbines and c) the velocity reduction caused by each array. Power, averaged over a tidal cycle, is normalised by the tuned power for a single isolated turbine in the same channel, highlighted by the dash-dot line in a) and b). Fitted curves highlight the trend with changing ε for n rows and crosses (pluses) show ε^{opt} for maximum $P^{tuned}_{per\ turb}$ for the 2-D (1-D) model.

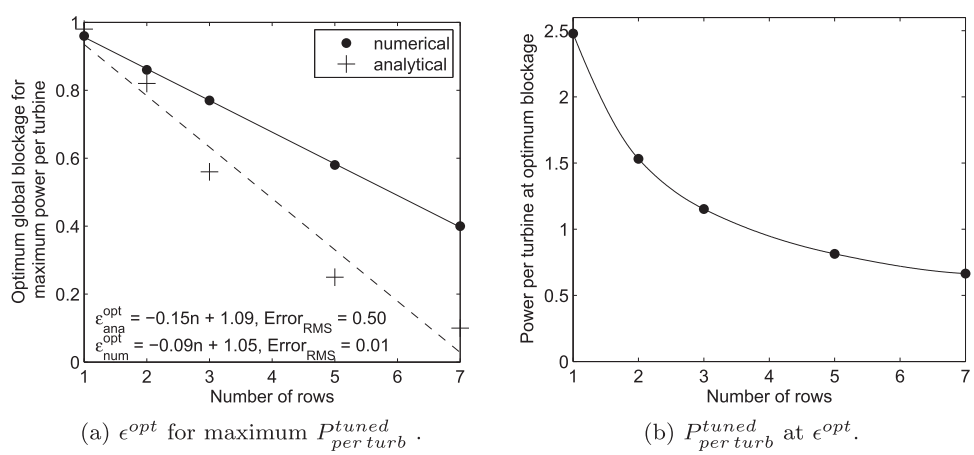


Fig. 3. Optimum global blockage and $P^{tuned}_{per\ turb}$ at that blockage for n rows and the linear fit to ε^{opt} as n increases.

The number of turbines is linked to ε by Equation (6) so the optimum number of turbines in each of n rows is given by

$$N^{opt} = (-0.10n + 1.06)\frac{W}{D}. \tag{10}$$

Care should be taken that this equations will not necessarily hold for all channels. The slope and offset will probably be different for channels of different sizes. It would be interesting to explore this relationship at higher n. However, the number of rows that can be simulated in this model at $10D$ spacing are physically constrained to 7 by the length of the channel. While the linear fit for 2-D numerical simulations is a remarkably good fit (RMS error = 0.01)

the linearity for the 1-D analytical results is not as good (RMS error = 0.5) due to the relatively lower ε^{opt} for 1, 5 and 7 rows in the 1-D model shown in 2a. The 1-D analytical slope is also steeper than the 2-D numerical slope. This highlights that while the 1-D analytical model is broadly sufficient, quantifying the processes involved in finding the optimum layout requires the extra resolution of the 2-D numerical model.

Comparing the power per turbine with the number of turbines in the array for different n highlights the difference in the power that each turbine captures for a single row compared to multiple rows (Fig. 2d). For 12 turbines are arranged in 3 rows (or 5 rows with 15 turbines), each turbine captures less than one third (one sixth) of the power that the same turbines capture when arranged in a single row. Further, to capture more total power than a single row of 12 turbines requires 20, 24, 40 or 49 turbines if they are grouped in 2, 3, 5 or 7 rows respectively.

3.3. Tuned total power

Unlike the power per turbine, the *total power* captured by an array continues to increase as either more turbines are added to each row or as more rows are added to the array (Fig. 2b). To capture more power from the flow, turbines can be added to each row or the number of rows can be increased. For instance 7 rows with $\varepsilon = 0.96$ capture 1.5× the power of 1 row with the same ε. This trend probably continues beyond the maximum number of rows simulated in this study as there is no evidence of a change in this trend for high n.

The *total* power continues to increase for higher global blockages, beyond the optimum global blockage for maximum power *per turbine* for an array with a given number of rows. The total power captured by an array with ε^{opt} changes with n, shown by the black crosses in Fig. 2b. As n increases from 1 to 3, the total power captured by the array of optimally blocked rows increases. However as n increases further from 3 to 7 the total power captured by optimally blocked rows decreases. Finally 7 rows at ε^{opt} capture only 0.75× the power that 1 optimally blocked row captures despite having almost 3× the turbines.

3.4. Impacts on free-stream velocity

As additional turbines are added to an array, whether it is by adding rows or by increasing the global blockage of each row, the free-stream velocity will be slowed by the extra force exerted on the fluid by the turbines. This reduction from natural velocity, without turbines, is the fractional change in free-stream velocity calculated over a tidal cycle by

$$\overline{R}_{free} = \frac{1}{T}\int_{T/2}^{3T/2} \frac{U_{free}(t)}{U_0(t)}\,dt. \tag{11}$$

$U_0(t)$ is the along stream velocity in the channel without turbines. $U_{free}(t)$ is the upstream, spatial average across the channel width (to remove turbulent length scales) of along stream velocity with the array impacting on the flow. The time average largely removes the effect of tidal frequency. Although there is a small phase difference between $U_0(t)$ and $U_{free}(t)$ due to turbines retarding the tidal cycle, this has minimal impact on $\overline{R}_{free}$.

The velocity slows towards Ref. [15]'s volume flow rate limit of 0.58 as the array size increases, Fig. 2c. This limit was derived from 1-D analytical simulations. The largest array simulated in the present, 2-D model does not quite reach this limit, slowing the flow to $\overline{R}_{free}^{tuned} = 0.61$.

Apart from very small and very large arrays ($\varepsilon < 0.3$ for 1 or 2 rows and $\varepsilon > 0.6$ for 2 or more rows) there is a linear decrease in $\overline{R}_{free}^{tuned}$ as the global blockage increases for an array with a given number of rows. This near-linear regime for mid-sized arrays was also seen in Fig. 2b suggesting there is a change in the channel-scale dynamics between small arrays, medium arrays in the near-linear regime and large arrays beyond this regime.

3.4.1. Total power capture increases linearly as free-stream velocity decreases

The total power that any uniform array will capture is linearly related to the reduction in normalised free-stream velocity by

$$P_{total}^{tuned} = -\kappa\overline{R}_{free} + \kappa, \tag{12}$$

shown by the linear trend in Fig. 4a. For the small channel $\kappa_{small} = 101$ and holds true, within a few percent, for any of the simulated arrays of 1–7 rows of 1–12 turbines. Further, the trend is consistent in arrays with 20 or 50 rows simulated analytically lending further support to this linearity. The slope and offset are different for the large strait where$\kappa_{large} = 810$, shown in Fig. 4b. In the large strait where inertia dominates friction there is a higher potential for power capture but this comes with a similar trend in flow reduction to the small channel.

The linearity is not a feature of a specific channel but is probably a general rule that could be applied to any channel. The general relationship between P_{total}^{tuned} and $\overline{R}_{free}$ in any given channel could be assessed with a single simulation or a calculation of the power potential of the channel containing a large array. Only the power capture and velocity reduction by a large array and the known point of zero power capture and zero velocity reduction in an empty channel are needed to find this linear relationship for a given channel.

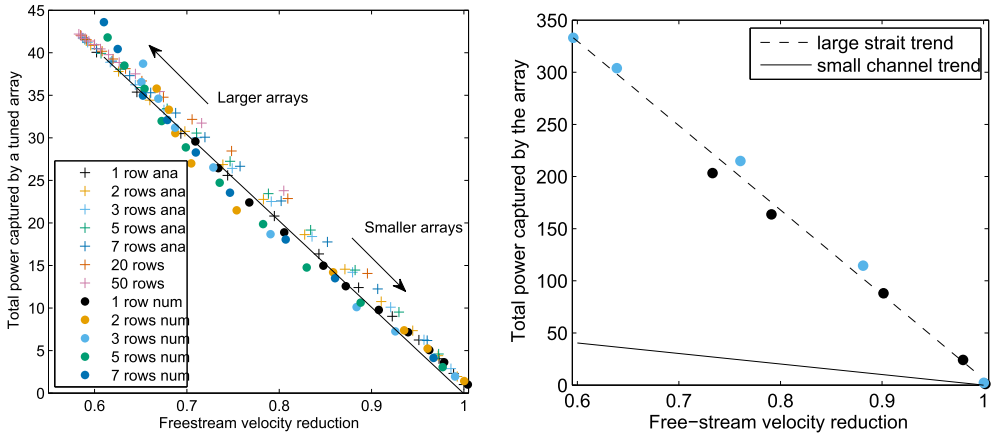
This linear relationship is a significant result as it means that building the array in a different layout does not change the flow reduction that will result from a given power capture. This is independent of global blockage, number of turbines in the array, and number of turbines per row. The turbines within the array of uniform rows must be tuned for maximum power for that layout for the linearity to hold.

3.5. Impacts on tidal head loss

For isolated turbines or even small arrays such as those in Fig. 1b and c, the local deviations in pressure do not significantly affect the channel-scale head loss. At high global blockages, e.g.: Fig. 1d and e, the turbines work together to each capture more power as the channel's tidal head loss and inertia in the flow force a large head loss over each row of turbines. This head loss over the row drives flow through turbines with significantly higher C_t^{tuned} than at lower blockages, Section 3.6. The head loss along the channel is held the same for all arrays by choice of boundary conditions while the free-stream velocity varies. This represents differences in the elevation of large water bodies at either end of the channel which are assumed to be too large to be affected by introducing an array to the channel.

There are four effects in the head loss around turbine arrays in a channel that help to understand the operation of optimally tuned arrays, evident in Fig. 5a and highlighted in Fig. 5b.

1. The driving head loss along the length of the channel, $\Delta p_{channel}$.
2. The total head loss over the array, Δp_{array}.
3. The average head loss over each row, $\Delta p_{row} = \Delta p_{array}/n$.



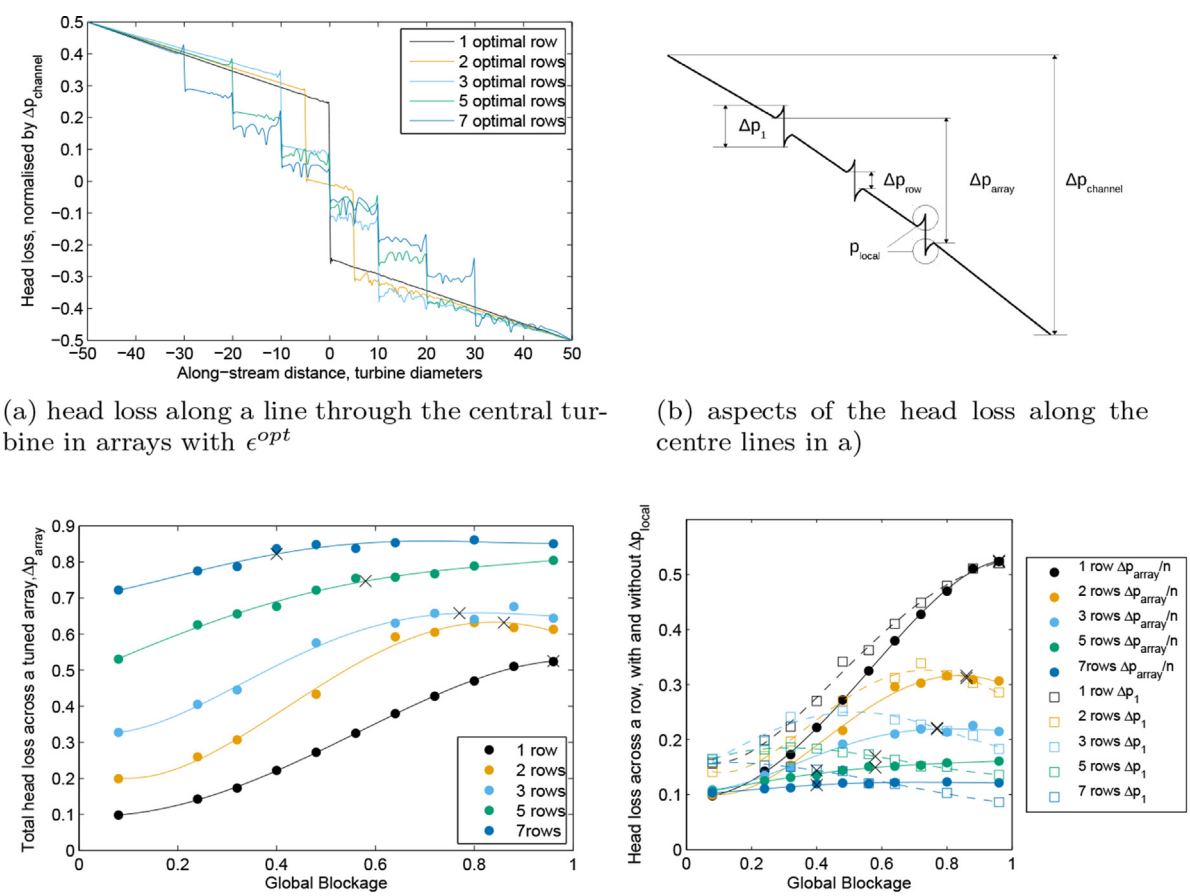
(a) small channel (b) large strait

Fig. 4. The total power captured by the flow by each array versus the reduction in free-stream velocity. Instantaneous free-stream velocity is normalised by the instantaneous velocity in an empty channel to remove the tidal frequency, then spatially averaged across an area of the channel and summed over the tidal period to remove turbulent scales and provide a single value for the tidally varying reduction in the free-stream velocity (11). Analytical results are new analysis of Vennell's 1-D analytical model. The linear fit is to 2-D results.

4. The local pressure effects associated with an expanding streamtube, Δp_{local}.

All head losses are expressed here as a fraction of the driving head loss along the length of the channel, $\Delta p/\Delta p_{channel}$.

Highly blocked rows of turbines exhibit two features of head loss that differ from that when rows have lower global blockage (Fig. 1b–e). First, there is a more pronounced, stepwise drop in Δp_{row} over each highly blocked row (Fig. 1d and e). This is in contrast to the larger change in Δp_{local} around rows with low global



(a) head loss along a line through the central turbine in arrays with ϵ^{opt}

(b) aspects of the head loss along the centre lines in a)

(c) head loss over the whole array

(d) head loss over a row of turbines, including and excluding Δp_{local}

Fig. 5. The instantaneous head loss over different parts of an array. Flow is to the right at the highest driving head difference, occurring at $t = T$. Head loss is a fraction of the total channel head loss. Black crosses are at ϵ^{opt}.

blockage, where the head loss does not spread to occupy the whole channel and is smaller (Fig. 1b and c). The pressure field around rows with only a few turbines is quite distinct, with each individual turbine having an individual effect on the flow.

In rows with low global blockage, the local pressure increases upstream and decreases downstream around individual turbines. These local changes arise from the local effect on the pressure field of an expanding streamtube in the wake behind turbines, in the same was as in actuator disk theory (e.g.: [27]). Here, the pressure changes result from 2-D representations of actuator disks with turbine scale wakes captured.

In contrast to the pressure field for a single turbine, at high global blockages these local pressure effects are not seen. In a highly blocked row there is little or no room for the streamtube downstream of a turbine to expand laterally, before the expansion of the streamtube is constrained by an expanding streamtube from an adjacent turbine. Without space for the streamtube to expand into, the pressure fields decrease in the stepwise manner seen and there is no increase in pressure upstream or decrease downstream, which would be expected for a single turbine in isolation.

These two effects on the head loss along the channel are particularly pronounced along a line through a central turbine (the solid black lines in Fig. 1b–e). Head loss along these lines for optimally blocked arrays of 1–7 rows of tuned turbines are shown in Fig. 5.

The gradient of head loss upstream and downstream of the array only changes slightly for tuned arrays at the optimal global blockage for a given number of rows (Fig. 1b). However, the head loss that occurs over a single row is almost as large as the combined head loss over 2 rows at the optimal global blockage for those 2 rows. The head loss over the whole optimally blocked array increases as the number of rows increases, simply because more rows occupy a larger proportion of the channel's length. However, there is also some change in the total head loss over the array due to a change in the gradient outside the array. These effects are explored further by considering the three aspects of the array head loss for all the array layouts simulated.

3.5.1. Total head loss over the array, Δp_{array}

Δp_{array} increases as the array gets larger, either through higher global blockage or more rows, up to 0.85 of the driving head loss for the largest arrays simulated, Fig. 5c. This total head loss is the difference in pressure between a single point upstream of the local pressure increase from the upstream turbine and the spatial mean in a region three diameters downstream of the turbine furthest downstream, removing the effect of Δp_{local}. The spatial mean reduces the impact of wake turbulence on this instantaneous head loss. Ref. [28] predicted that a large array, capturing a significant fraction of a channel's potential, will cause nearly all of the channel's head loss to occur over the array. This is comparable with the increasing proportion of the channel's head lost over the array shown in Fig. 5c. There is no reason to expect that the head lost would not continue to increase towards 1 for even larger arrays than those simulated here.

For up to 3 rows there is a global blockage that gives a maximum highest head loss that is very close to ε^{opt}, indicated by crosses in Fig. 5c. For arrays with 5 or 7 rows, Δp_{array} increases continuously as more turbines are added to each row. This continuous increase in the head loss as more tuned turbines are added to a large array of many rows shows that adding turbines to each row lowers the pressure gradient in the region of the channel outside the array. This occurs as an increasing fraction of the driving head loss is used to force flow through the turbines and the free-stream velocity is consequently reduced.

3.5.2. Head loss over each row, Δp_{row}

The shape of the curves for this average fractional head loss over each row (solid lines and dots in Fig. 5d) is remarkably similar to the shape of each curve for power per turbine in Fig. 2a. This suggests that the head loss over each row of tuned turbines is fundamentally the driving dynamic that determines the power that the array captures from the flow. The only deviation from the similarity between the head loss per row and the power per turbine is for 5 or more highly blocked rows. For these large arrays, the head loss does not decrease, whereas the power per turbine does decrease, at $\varepsilon > \varepsilon^{opt}$. ε^{opt} is close to, but slightly higher than, the ε that results in the highest head loss over each row of turbines, for 1 to 3 rows.

3.5.3. Local effects around a single row, Δp_1

The differences between the average head loss over each row (solid lines) and the head loss over the most upstream row (dashed lines) in Fig. 5d are mostly in the head loss over rows with low global blockage. It is only for low global blockages, where there is lateral space between turbines for streamtube expansion in the wake, that the local pressure increase upstream and loss downstream occurs.

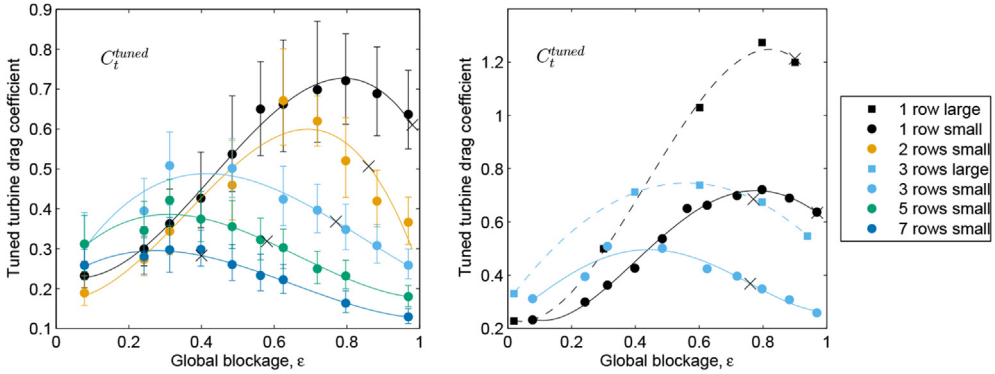
3.6. Optimum tuning for maximum power capture in each array

Turbines in each layout must be tuned to find the tuned turbine drag coefficient for maximum power capture for each specific layout, C_t^{tuned}. The tuning error bars in Fig. 6a indicates the range of turbine drag coefficients that give power to within 99.5% of the maximum power at C_t^{tuned}. The tuning range is quite high, in some cases covering almost half the extent of the curve. Despite this large range the fitted curves still highlight trends in the expected turbine tuning that lie outside the tuning range. This turbine tuning has been used for all of the previous analysis of array layout optimisation. It is crucial to perform this first step of tuning turbines within an array layout for maximum power capture by that specific layout. This is because the turbine drag tuning varies by a factor of up to six amongst arrays of different layouts (Fig. 6a). Decreasing C_t to below C_t^{tuned} for a specific layout reduces power capture directly through Equation (7). Increasing C_t above C_t^{tuned} decreases power capture by reducing the free-stream flow and the flow within the turbines.

The optimum tuning for a given array layout results from an interaction of reduced free-stream velocity decreasing the power available for turbines to capture and increased head loss over each turbine increasing the power available. For an array with a given number of rows and low ε, C_t^{tuned} initially increases as ε increases. As ε increases in this regime, Δp_{row} increases allowing more power to be captured despite the reduction in free-stream velocity due to higher C_t and more turbines. At ε a little below ε^{opt} the C_t^{tuned} reaches a maximum. Beyond this maximum, C_t^{tuned} must be lower than at the maximum to capture maximum power from that layout. In this regime, further increases in C_t would slow the free-stream velocity to a point beyond optimal flow reduction for maximum power and the increased head loss cannot compensate.

Generally, in arrays with more rows the individual turbines need to be tuned to a lower C_t to get the most power out of the array (Fig. 6a). Apart from very small arrays of 1 or 2 rows with $\varepsilon < 0.5$, C_t^{tuned} decreases as the number of rows in the array increases, for all global blockages. This relates to each turbine in a larger array being detuned to capture less power so that the overall array captures the most power.

For an array with a given number of rows the maximum value of C_t^{tuned} is highest for arrays constructed of 1 row of turbines and decreases down to a lower drag coefficient for 7 rows. For large arrays C_t^{tuned} can be as low as one sixth of C_t^{tuned} for a single



(a) small channel. (b) large strait and small channel

Fig. 6. The optimum turbine drag coefficient for maximum power capture, C_t^{tuned}, in each specific array layout in the a) small channel b) small channel and large strait. Crosses show ε^{opt}. Tuning bars in a) show the range of C_t to give power to within 0.995 of P^{tuned}.

highly blocked row. A factor of six is a very significant difference in such a fundamental design specification and shows just how important the tuning of turbines within each layout is to the power capture.

3.6.1. Optimum tuning in the large strait

Tuning of C_t for maximum power capture from a given layout follows similar patterns, with varying global blockage, in the large strait compared to the small channel, Fig. 6b. C_t^{tuned} is up to 1.7× higher in the large strait than in the small channel for all arrays except a single turbine. This difference is larger for arrays with higher global blockage than low global blockage, possibly largely due to the deeper water in the large strait. The higher inertia in the large strait allows a higher C_t^{tuned} at a given global blockage relative to the small, friction-dominated channel. This indicates that turbine tuning is specific to not only array layout but also the dynamics of the channel in which the array is deployed.

3.7. Velocity and head loss in optimised layouts

As the number of rows in the array increases from 1 to 7, the optimum global blockage decreases, as shown by the optimum layouts for maximum power per turbine in Fig. 7a–e. These optimum layouts show the effect of designing arrays for both C_t^{tuned} and ε^{opt} for a given number of rows. In the optimised array with 7 rows C_t^{tuned} is half the C_t^{tuned} for the optimum single row (Fig. 6a). Turbines in a larger array need to be individually detuned, with lower C_t^{tuned}, as the head loss driving the flow through the array is spread over more rows of turbines. At the same time the free-stream velocity in optimised arrays only increases a little, from $0.72U_0$ for 1 row to $0.76U_0$ for 7 rows. The stepped head loss in these optimised arrays is evenly distributed along the rows in the array (Fig. 5a) providing the optimum flow for power capture by an array with each number of rows. It is this combination of head loss and tuned velocity reduction that determines the optimum global blockage for an array with a given number of rows.

4. Discussion

Optimum layout is a balance between total power captured and power per turbine. There is an optimum global blockage for the maximum power *per turbine*. However the *total* power captured by the array continues to increase as turbines are added to each row. At global blockages higher than ε^{opt} there is a diminishing return in power per turbine for adding extra turbines to a row as extra turbines slow the flow in the channel to the point where there is less power available to be captured by the array. This does not decrease the total power captured by the array due to the extra turbines in larger arrays it just decreases the power capture of each turbine, in agreement with Vennell [8] and Adcock [16].

A tidal energy array developer must assess the optimum layout for a given development based on larger scale economics of the project. A site developer might decide that the total power captured is more important to the economics of operation than the marginal power gain from each added turbine, based on fixed costs that might include expensive transmission lines, for example. However, this decision must be made with the awareness that adding more rows severely limits the power that each turbine can capture. This drop in the power available to each turbine is due to the reduction in the velocity through the channel and the change in head loss over the array. It is not just flow reduction from upstream wakes that limit the power capture.

Previous studies of array optimisation have focussed exclusively on the wake interaction. Neither the 2-D numerical or 1-D analytical models presented here captures the full 3-D dynamical wake. However, the wake dynamics appear to be the dominant effect *only* for small arrays of a few turbines per row. Further, previous studies of array optimisation have driven the flow through an array by an imposed velocity. However, for large arrays this assumption of constant velocity is not valid and leads to incorrect conclusions about power capture and optimum layout. In larger arrays, each row of turbines blocks the channel so that a step-like head loss develops. In these larger arrays the head loss and free-stream flow reduction appear to be more significant than wake interaction as the velocity deficit region shortens considerably in downstream rows.

The turbine tuning required to achieve maximum power extraction from a given layout varies from highest for 1 highly blocked row to $1/6^{th}$ of that value for 7 highly blocked rows. Turbine manufacturers and site developers should be aware of this variation in C_t^{tuned} for different array layouts. This large variation in the design specification for turbines built for use in different arrays means that turbines designed and tuned for use in isolated test operation will be far from optimally tuned for extracting maximum power in large array deployments.

Both the 2-D numerical and 1-D analytical model assume "small" Froude numbers, ignoring potential for free surface effects around the turbines, specifically the potential for hydraulic jumps. Even in gaps between turbines (where the highest velocities occur and the smallest length scales exist) the Froude numbers are significantly less than 0.3 so this assumption is reasonable.

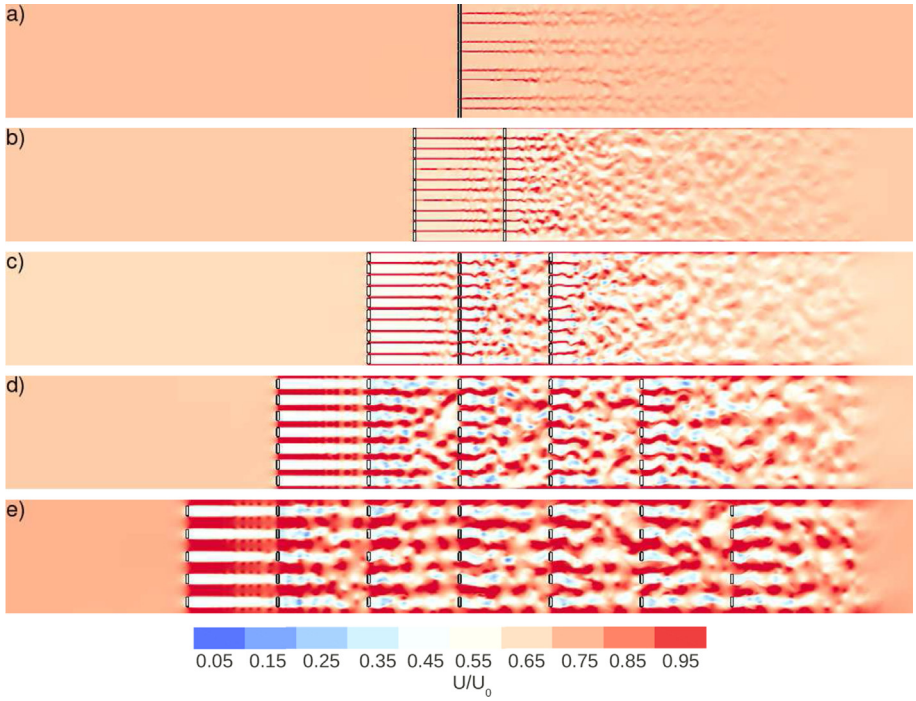


Fig. 7. Along-stream velocity snapshots at maximum rightward flow for arrays with optimal global blockage in 1–7 uniform rows of tuned turbines in a) to e), respectively.

4.1. Further work

Ideally these simulations would be done in 3-D but the simulation time required to simulate the number of arrays and tunings presented in the current paper would be prohibitive. Hence this paper is part of a staged approach to understand the optimum layout of tidal arrays: 1) Vennell's analytical model 2) 2-D simulations presented in this paper guided by Vennell's results, 3) arrays with variations away from uniform row layout simulated in 2D, 4) simulations in 3D should be made using the knowledge gained in all previous steps.

If these arrays were simulated in 3-D, the dissipative mixing of the vertically bypassing flow with the wake would increase, probably shortening the wakes and possibly reducing the importance of horizontal blockage. An analytical model that matches the momentum loss and power capture in 2-D numerical models to those in 3-D numerical models by applying a correction to the power capture has recently been developed by Vogel et al. [29], suggesting that results of the present study may be reduced by up to 20%. It should be noted however that the present study does not 'smear' turbines over a group of cells around the turbine location as was assumed by Vogel et al.

Multiple tidal constituents [30] and turbulent upstream flow [31,32] would further complicate micro arrangement of array layout optimisation. It would be interesting to test a subset of the models developed in this paper at a real channel site to explore which effects dominate for a combination of local bathymetry and blockages. Other site geometries such as the laterally non-bounded, non-channel sites studied by Ref. [33] or channels connecting an estuary to the ocean [34] have not been considered in this study of arrays in channels but could be explored with a simple modification to this model.

5. Conclusion

Optimum array layout for tidal turbines in channels is a compromise between the total power captured by the whole array, the power that each turbine captures and the effect on free-stream velocity. In agreement with Vennell [8] and Adcock [16] the optimum layout for maximum power per turbine is a single, nearly totally blocked row, demonstrating that adding rows significantly decreases the capability of each turbine to capture power as the channel-scale flow is slowed by adding rows. Adding more rows to an array increases the total power captured simply by increasing the number of turbines. However, adding rows decreases the power that each turbine captures as the free-stream velocity is slowed by the extra drag. For a given number of rows there is an optimum global blockage for maximum power per turbine, ε^{opt}. Adding more turbines, beyond ε^{opt}, increases P_{total}, but decreases $P_{per\ turb}$. A significant new result is that this ε^{opt} decreases linearly from 1.0 (completely blocked) for 1 row, to 0.4 (less than half blocked) for 7 rows.

The most significant result demonstrated by the present modelling is the linear relationship between the total power captured by an optimally-tuned array and the resulting free-stream velocity reduction. This suggests that the simplifications inherent in 1-D analytical models such as no individual turbines and not explicitly resolving the wake are sufficient. As a consequence, the extra processes contained in the numerical model are secondary at least in the confined channel situations. The linear relationship shows there is an unchanging balance between the global environmental benefit of generating sustainable electricity and the local environmental cost of velocity reduction in the channel that cannot be exceeded with uniform rows of tidal turbines. Velocity reduction will influence sediment deposition/erosion as well as influence habitat and hence local ecosystem function. Quantifying these processes does require the extra resolution provided by the numerical approach. This improved understanding will enhance confidence in obtaining maximum power capture, optimum layout and tuning while also providing a basis for improved environmental impact assessment.

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References

[1] C. Garrett, P. Cummins, The efficiency of a turbine in a tidal channel, J. Fluid Mech. 588 (-1) (2007) 243–251, http://dx.doi.org/10.1017/S0022112007007781.
[2] T. Nishino, R.H.J. Willden, The efficiency of an array of tidal turbines partially blocking a wide channel, J. Fluid Mech. 708 (2012) 596–606, http://dx.doi.org/10.1017/jfm.2012.349.
[3] W.M.J. Batten, A.S. Bahaj, CFD simulation of a small farm of horizontal axis marine current turbines, in: A.A.M. Sayigh (Ed.), Proc. World Renew. Energy Congr. (WREC-IX), Florence, Italy 19–25 August 2006, Elsevier Science, 2006.
[4] A.J. MacLeod, S. Barnes, K.G. Rados, I.G. Bryden, Wake effects in tidal current turbine farms, in: Int. Conf. Mar. Renew. Energy-conference Proc., 2002, pp. 49–53.
[5] W.M.J. Batten, A.S. Bahaj, A.F. Molland, J.R. Chaplin, Experimentally validated numerical method for the hydrodynamic design of horizontal axis tidal turbines, Ocean Eng. 34 (7) (2007) 1013–1020, http://dx.doi.org/10.1016/j.oceaneng.2006.04.008.
[6] S.H. Lee, S.H. Lee, K. Jang, J. Lee, N. Hur, A numerical study for the optimal arrangement of ocean current turbine generators in the ocean current power parks, Curr. Appl. Phys. 10 (2, Suppl. 1) (2010) S137–S141, http://dx.doi.org/10.1016/j.cap.2009.11.018.
[7] M.J. Churchfield, Y. Li, P.J. Moriarty, A large-eddy simulation study of wake propagation and power production in an array of tidal-current turbines, Philos. Trans. R. Soc. A Math. Eng. Sci. 371 (1985), http://dx.doi.org/10.1098/rsta.2012.0421.
[8] R. Vennell, Tuning turbines in a tidal channel, J. Fluid Mech. 663 (2010) 253–267, http://dx.doi.org/10.1017/S0022112010003502.
[9] R. Vennell, S.W. Funke, S. Draper, C. Stevens, T. Divett, Designing large arrays of tidal turbines: a synthesis and review, Renew. Sustain. Energy Rev. 41 (0) (2015) 454–472, http://dx.doi.org/10.1016/j.rser.2014.08.022. URL, http://www.sciencedirect.com/science/article/pii/S1364032114006984.
[10] T. Divett, R. Vennell, C. Stevens, Optimization of multiple turbine arrays in a channel with tidally reversing flow by numerical modelling with adaptive mesh, Philos. Trans. R. Soc. A Math. Eng. Sci. 371 (1985), http://dx.doi.org/10.1098/rsta.2012.0251.
[11] T.A. Divett, Optimising Design of Large Tidal Energy Arrays in Channels: Layout and Turbine Tuning for Maximum Power Capture Using Large Eddy Simulations with Adaptive Mesh, PhD thesis, University of Otago, New Zealand, Sep. 2014. URL, http://otago.ourarchive.ac.nz/handle/10523/4995.
[12] R. Vennell, Tuning tidal turbines in-concert to maximise farm efficiency, J. Fluid Mech. 671 (2011) 587–604, http://dx.doi.org/10.1017/S0022112010006191.
[13] R. Vennell, The energetics of large tidal turbine arrays, Renew. Energy 48 (2012) 210–219, http://dx.doi.org/10.1016/j.renene.2012.04.018. URL, http://www.sciencedirect.com/science/article/pii/S0960148112002595.
[14] R. Vennell, Realizing the potential of tidal currents and the efficiency of turbine farms in a channel, Renew. Energy 47 (0) (2012) 95–102, http://dx.doi.org/10.1016/j.renene.2012.03.036.
[15] C. Garrett, P. Cummins, The power potential of tidal currents in channels, Proc. R. Soc. A Math. Phys. Eng. Sci. 461 (2060) (2005) 2563–2572, http://dx.doi.org/10.1098/rspa.2005.1494.
[16] T. A. Adcock, S. Draper, G. T. Houlsby, A. G. Borthwick, S. Serhadlioglu, The available power from tidal stream turbines in the Pentland Firth, Proc. R. Soc. A Math. Phys. Eng. Sci. 469 (2157)
[17] T. Nishino, R.H. Willden, Effects of 3-D channel blockage and turbulent wake mixing on the limit of power extraction by tidal turbines, Int. J. Heat Fluid Flow 37 (null) (2012) 123–135, http://dx.doi.org/10.1016/j.ijheatfluidflow.2012.05.002.
[18] S. Popinet, Gerris: a tree-based adaptive solver for the incompressible Euler equations in complex geometries, J. Comput. Phys. 190 (2) (2003) 572–600.
[19] S. Popinet, G. Rickard, A tree-based solver for adaptive ocean modelling, Ocean Model. 16 (3–4) (2007) 224–249, http://dx.doi.org/10.1016/j.ocemod.2006.10.002.
[20] S. Popinet, M. Smith, C. Stevens, Experimental and numerical study of the turbulence characteristics of air flow around a research vessel, J. Ocean Atmos. Tech. 21 (10) (2004) 1574–1589.
[21] S. Popinet, Adaptive modelling of long-distance wave propagation and fine-scale flooding during the Tohoku tsunami, Nat. Hazards Earth Syst. Sci. 12 (4) (2012) 1213–1227, http://dx.doi.org/10.5194/nhess-12-1213-2012. URL, http://gfs.sf.net/papers/popinet2012.pdf.
[22] J.P. Boris, F.F. Grinstein, E.S. Oran, R.L. Kolbe, New insights into large eddy simulation, Fluid Dyn. Res. 10 (4) (1992) 199–228.
[23] D.H. Porter, A. Pouquet, P.R. Woodward, Kolmogorov-like spectra in decaying three-dimensional supersonic flows, Phys. Fluids 6 (1994) 2133.
[24] T. Divett, R. Vennell and C. Stevens, Channel scale optimisation of large tidal turbine arrays in packed rows using large eddy simulations with adaptive mesh, In: *Proc. 2nd Asian Wave Tidal Energy Conf.*, 2014, Tokyo.
[25] R. Vennell, Exceeding the Betz limit with tidal turbines, Renew. Energy 55 (2013) 277–285.
[26] D.R. Plew, C.L. Stevens, Numerical modelling of the effect of turbines on currents in a tidal channel-Tory channel, New Zealand, Renew. Energy 57 (0) (2013) 269–282.
[27] T. Burton, D. Sharpe, N. Jenkins, E. Bossanyi, Wind Energy Handbook, URL, John Wiley & Sons, Ltd, Chichester, UK, 2002, http://search.ebscohost.com.ezproxy.otago.ac.nz/login.aspx%3fdirect=true%5c&db=iih%5c&jid=VPN%5c&site=ehost-live.
[28] S. Couch, I. Bryden, Tidal current energy extraction: hydrodynamic resource characteristics, Proc. Inst. Mech. Eng. Part M J. Eng. Marit. Environ. 220 (4) (2006) 185–194, http://dx.doi.org/10.1243/14750902JEME50.
[29] C. Vogel, R. Willden, G. Houlsby, A correction for depth-averaged simulations of tidal turbine arrays, in: In Proc. 10th European Wave and Tidal Energy Conference (EWTEC), 2013. Aalborg, Denmark.
[30] T. Divett, R. Vennell, C. Stevens, Maximising energy capture by fixed orientation tidal stream turbines in time-varying off axis current, in: Proc. Australas. Coasts Ports Conf. Sept. 2009, 2009. Wellington, New Zeal.
[31] C.L. Stevens, M.J. Smith, B. Grant, C.L. Stewart, T. Divett, Tidal energy resource complexity in a large strait: the Karori Rip, Cook Strait, Cont. Shelf Res. 33 (0) (2012) 100–109, http://dx.doi.org/10.1016/j.csr.2011.11.012. URL, http://www.sciencedirect.com/science/article/pii/S027843431100358X.
[32] M.E. Harrison, W.M.J. Batten, L.E. Myers, A.S. Bahaj, Comparison between CFD simulations and experiments for predicting the far wake of horizontal axis tidal turbines, Renew. Power Gener. IET 4 (6) (2010) 613–627, http://dx.doi.org/10.1049/iet-rpg.2009.0193.
[33] S. Draper, A.G.L. Borthwick, G.T. Houlsby, Energy potential of a tidal fence deployed near a coastal headland, Philos. Trans. R. Soc. A Math. Phys. Eng. Sci. 371 (1985) (2013), http://dx.doi.org/10.1098/rsta.2012.0176, 20120176, http://www.ncbi.nlm.nih.gov/pubmed/23319703. URL.
[34] R.H. Karsten, J.M. McMillan, M.J. Lickley, R.D. Haynes, Assessment of tidal current energy in the Minas Passage, Bay of Fundy, Proc. Inst. Mech. Eng. Part A J. Power Energy 222 (5) (2008) 493–507, http://dx.doi.org/10.1243/09576509JPE555.